

SmoothWall FAQ and Troubleshooting Guide

or

The book of serious answers to questions you should NEVER now need to ask the SmoothWall team

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Preface by: Eric S. Raymond

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Preface

How To Ask Questions The Smart Way
by Eric S. Raymond
Author of ‘Cathedral and the Bazaar’

Introduction

In the world of hackers, the kind of answers you get to your technical questions depends as much on the way you ask the questions as on the difficulty of developing the answer. This guide will teach you how to ask questions in a way that is likely to get you a satisfactory answer.

The first thing to understand is that hackers actually like hard problems and good, thought-provoking questions about them. If we didn't, we wouldn't be here. If you give us an interesting question to chew on we'll be grateful to you; good questions are a stimulus and a gift. Good questions help us develop our understanding, and often reveal problems we might not have noticed or thought about otherwise. Among hackers, "Good question!" is a strong and sincere compliment.

Despite this, hackers have a reputation for meeting simple questions with what looks like hostility or arrogance. It sometimes looks like we're reflexively rude to newbies and the ignorant. But this isn't really true.

What we are, unapologetically, is hostile to people who seem to be unwilling to think or do their own homework before asking questions. People like that are time sinks -- they take without giving back, they waste time we could have spent on another question more interesting and another

person more worthy of an answer. We call people like this "losers" (and for historical reasons we sometimes spell it "lusers").

We realise that there are many people who just want to use the software we write, and have no interest in learning technical details. For most people, a computer is merely a tool, a means to an end; they have more important things to do and lives to live. We acknowledge that, and don't expect everyone to take an interest in the technical matters that fascinate us. Nevertheless, our style of answering questions is tuned for people who do take such an interest and are willing to be active participants in problem-solving. That's not going to change. Nor should it; if it did, we would become less effective at the things we do best.

We're (largely) volunteers. We take time out of busy lives to answer questions, and at times we're overwhelmed with them. So we filter ruthlessly. In particular, we throw away questions from people who appear to be losers in order to spend our question-answering time more efficiently, on winners.

If you find this attitude obnoxious, condescending, or arrogant, check your assumptions. We're not asking you to genuflect to us; in fact, most of us would love nothing more than to deal with you as an equal, if you put in the effort required to make that possible. If you can't live with this sort of discrimination, we suggest you pay somebody for a commercial support contract instead of asking hackers to personally donate help to you.

If you decide to come to us for help, you don't want to be one of the losers. You don't want to seem like one, either. The best way to get a rapid and responsive answer is to ask it

like a winner; to ask it like a person with smarts, confidence, and clues who just happens to need help on one particular problem.

Before You Ask

Before asking a technical question by email, or in a newsgroup, or on a web site chat board, do the following:

1. Try to find an answer by reading the manual.
2. Try to find an answer by reading a FAQ.
3. Try to find an answer by searching the Web.
4. Try to find an answer by asking a skilled friend.

When you ask your question, display the fact that you have done these things first; this will help establish that you're not being a lazy sponge and wasting peoples' time. Better yet, display what you have learned from doing these things. We like answering questions for people who have demonstrated that they can learn from the answers.

Prepare your question. Think it through. Hasty-sounding questions get hasty answers, or none at all. The more you do to demonstrate that you have put thought and effort into solving your problem before asking for help, the more likely you are to actually get help.

Beware of asking the wrong question. If you ask one that is based on faulty assumptions, J. Random Hacker is quite likely to reply with a uselessly literal answer while thinking "Stupid question...", and hoping that the experience of getting what you asked for rather than what you needed will teach you a lesson.

Never assume you are entitled to an answer. You are not; you aren't, after all, paying for the service. You will earn an answer, if you earn it, by asking a question that is substantial, interesting, and thought-provoking, one that implicitly contributes to the experience of the community rather than merely passively demanding knowledge from others.

On the other hand, making it clear that you are able and willing to help in the process of developing the solution is a very good start. "Can someone provide a pointer?", "What is my example missing?" and "Is there a site I should have checked?" are more likely to get answered than "Please post the exact procedure I should use." because you're making it clear that you're truly willing to complete the process if someone can simply point you in the right direction.

When You Ask

Choose your forum carefully

Be sensitive in choosing where you ask your question. You are likely to be ignored, or written off as a loser, if you:

- post your question to a forum where it is off topic
- post a very elementary question to a forum where advanced technical questions are expected, or vice-versa
- cross-post to too many different newsgroups

Hackers blow off questions that are inappropriately targeted in order to try to protect their communications channels from being drowned in irrelevance. You don't want this to happen to you.

In general, questions to a well-selected public forum are more likely to get useful answers than equivalent questions to a private one. There are multiple reasons for this. One is simply the size of the pool of potential respondents. Another is the size of the audience; hackers would rather answer questions that educate a lot of people than questions, which only serve a few.

Write in clear, grammatical, correctly-spelled language

We've found by experience that people who are careless and sloppy writers are usually also careless and sloppy at thinking and coding (often enough to bet on, anyway). Answering questions for careless and sloppy thinkers is not rewarding; we'd rather spend our time elsewhere.

So expressing your question clearly and well is important. If you can't be bothered to do that, we can't be bothered to pay attention. Spend the extra effort to polish your language. It doesn't have to be stiff or formal - in fact, hacker culture values informal, slangy and humorous language used with precision. But it has to be precise; there has to be some indication that you're thinking and paying attention.

Spell correctly. Don't confuse "its" with "it's" or "loose" with "lose". Don't TYPE IN ALL CAPS, this is read as shouting and considered rude. If you write like a semi-literate boob, you will probably be ignored. Writing like a l33t script kiddie hax0r is the absolute kiss of death and guarantees you will receive nothing but stony silence (or, at best, a heaping helping of scorn and sarcasm) in return.

If you are asking questions in a forum that does not use your native language, you will get a limited amount of slack for spelling and grammar errors, but no extra slack at all for

laziness (and yes, we can usually spot that difference). Also, unless you know what your respondent's languages are, write in English. Busy hackers tend to simply flush questions in languages they don't understand, and English is the working language of the net. By writing in English you minimise your chances that your question will be discarded unread.

Send questions in formats that are easy to understand

If you make your question artificially hard to read, it is more likely to be passed over in favour of one that isn't. So:

- Send plain text mail, not HTML.
- Don't send mail in which entire paragraphs are single multiply-wrapped lines. (This makes it too difficult to reply to just part of the message.)
- Don't send MIME Quoted-Printable encoding either; all those =20 glyphs scattered through the text are ugly and distracting.
- Never, ever expect hackers to be able to read closed proprietary document formats like Microsoft Word. Most hackers react to these about as well as you would to having a pile of steaming pig manure dumped on your doorstep.
- If you're sending mail from a Windows machine, turn off Microsoft's stupid "Smart Quotes" feature. This is so you avoid sprinkling garbage characters through your mail.

Use meaningful, specific subject headers

On mailing lists or newsgroups, the subject header is your golden opportunity to attract qualified experts' attention in

around 50 characters or fewer. Don't waste it on babble like "Please help me" (let alone "PLEASE HELP ME!!!!"). Don't try to impress us with the depth of your anguish; use the space for a super-concise problem description instead.

Stupid:

> HELP! Video doesn't work properly on my laptop!

Smart:

> XFree86 4.1 misshapen mouse cursor, Fooware MV1005 vid. chipset

Be precise and informative about your problem

- Describe the symptoms of your problem or bug carefully and clearly.
- Describe the environment in which it occurs (machine, OS, application, whatever).
- Describe the research you did to try and understand the problem before you asked the question.
- Describe the diagnostic steps you took to try and pin down the problem yourself before you asked the question.
- Describe any recent changes in your computer or software configuration that might be relevant.

Do the best you can to anticipate the questions a hacker will ask, and to answer them in advance in your request for help.

Simon Tatham has written an excellent essay entitled How to Report Bugs Effectively which can be found at http://www.chiark.greenend.org.uk/~sgtatham/bugs.html. I strongly recommend that you read it.

Volume is not precision

You need to be precise and informative. This end is not served by simply dumping huge volumes of code or data into a help request. If you have a large, complicated test case that is breaking a program, try to trim it and make it as small as possible.

This is useful for at least three reasons. One: being seen to invest effort in simplifying the question makes it more likely that you'll get an answer, Two: simplifying the question makes it more likely you'll get a useful answer. Three: In the process of refining your bug report, you may develop a fix or workaround yourself.

Describe the problem's symptoms, not your guesses

It's not useful to tell hackers what you think is causing your problem. (If your diagnostic theories were such hot stuff, would you be consulting others for help?) So, make sure you're telling them the raw symptoms of what goes wrong, rather than your interpretations and theories. Let them do the interpretation and diagnosis.

Stupid:

> I'm getting back-to-back SIG11 errors on kernel compiles, and suspect a hairline crack on one of the motherboard traces. What's the best way to check for those?

Smart:

> My home-built K6/233 on an FIC-PA2007 motherboard (VIA Apollo VP2 chipset) with 256MB Corsair PC133 SDRAM starts getting frequent SIG11 errors about 20 minutes after power-on during the

course of kernel compiles, but never in the first 20 minutes. Rebooting doesn't restart the clock, but powering down overnight does. Swapping out all RAM didn't help. The relevant part of a typical compile session log follows.

Describe your problem's symptoms in chronological order

The most useful clues in figuring out something that went wrong often lie in the events immediately prior. So, your account should describe precisely what you did, and what the machine did, leading up to the blow up. In the case of command-line processes, having a session log (e.g., using the script utility) and quoting the relevant twenty or so lines is very useful.

If the program that blew up on you has diagnostic options (such as -v for verbose), try to think carefully about selecting options that will add useful debugging information to the transcript.

If your account ends up being long (more than about four paragraphs), it might be useful to succinctly state the problem up top, then follow with the chronological tale. That way, hackers will know what to watch for in reading your account.

Don't ask people to reply by private email

Hackers believe solving problems should be a public, transparent process during which a first try at an answer can and should be corrected if someone more knowledgeable notices that it is incomplete or incorrect. Also, they get some

of their reward for being respondents from being seen to be competent and knowledgeable by their peers.
When you ask for a private reply, you are disrupting both the process and the reward. Don't do this. It's the respondent's choice whether to reply privately, and if he does, it's usually because he thinks the question is too obvious or ill formed to be interesting to others.

There is one limited exception to this rule. If you think the question is such that you are likely to get a lot of answers that are all pretty similar, then the magic words are "email me and I'll summarise the answers for the group". It is courteous to try and save the mailing list or newsgroup a flood of substantially identical postings, but you have to keep the promise to summarise.

Prune pointless queries

Resist the temptation to close your request for help with semantically-null questions like "Can anyone help me?" or "Is there an answer?" First: if you've written your problem description halfway competently, such tacked-on questions are at best superfluous. Second: because they are superfluous, hackers find them annoying, and are likely to return logically impeccable but dismissive answers like "Yes, you can be helped" and "No, there is no help for you."

Courtesy never hurts, and sometimes helps

Be courteous. Use "Please" and "Thanks in advance". Make it clear that you appreciate the time people spend helping you for free.

To be honest, this isn't as important as (and cannot substitute for) being grammatical, clear, precise and descriptive,

avoiding proprietary formats etc.; hackers in general would rather get somewhat brusque but technically sharp bug reports than polite vagueness. (If this puzzles you, remember that we value a question by what it teaches us.)

However, if you've got your technical ducks in a row, politeness does increase your chances of getting a useful answer.

Follow up with a brief note on the solution

Send a note after the problem has been solved to all who helped you; let them know how it came out and thank them again for their help. If the problem attracted general interest in a mailing list or newsgroup, it's appropriate to post the follow-up there.

Your follow-up doesn't have to be long and involved; a simple "Howdy - it was a failed network cable! Thanks, everyone. - Bill" would be better than nothing. In fact, a short and sweet summary is better than a long dissertation unless the solution has real technical depth.

Besides being courteous and informative, this sort of follow-up helps everybody who assisted feel a satisfying sense of closure about the problem. If you are not a techie or hacker yourself, trust us that this feeling is very important to the gurus and experts you tapped for help. Problem narratives that trail off into unresolved nothingness are frustrating things; hackers itch to see them resolved. The good karma that scratching that itch earns you will be very, very helpful to you next time you need to pose a question.

How to Interpret Answers

RTFM and STFW: How To Tell You've Seriously Screwed Up

There is an ancient and hallowed tradition: if you get a reply that reads "RTFM", the person who sent it thinks you should have Read The Fucking Manual. He is almost certainly right. Go read it.

RTFM has a younger relative. If you get a reply that reads "STFW", the person who sent it thinks you should have Searched The Fucking Web. He is almost certainly right. Go search it.

Often, the person sending either of these replies has the manual or the web page with the information you need open, and is looking at it as he types. These replies mean that he thinks (a) the information you need is easy to find, and (b) you will learn more if you seek out the information than if you have it spoon-fed to you.

You shouldn't be offended by this; by hacker standards, he is showing you a rough kind of respect simply by not ignoring you. You should instead thank him for his grandmotherly kindness.

If you don't understand...

If you don't understand the answer, do not immediately bounce back a demand for clarification. Use the same tools that you used to try and answer your original question (manuals, FAQs, the Web, skilled friends) to understand the answer. If you need to ask for clarification, exhibit what you have learned.

For example, suppose I tell you: "It sounds like you've got a stuck zentry; you'll need to clear it." Then:

Here's a bad follow-up question:

"What's a zentry?"

Here's a good follow up question:

"OK, I read the man page and zentries are only mentioned under the -z and -p switches. Neither of them says anything about clearing zentries. Is it one of these or am I missing something here?"

Dealing with rudeness

Much of what looks like rudeness in hacker circles is not intended to give offence. Rather, it's the product of the direct, cut-through-the-bullshit communications style that is natural to people who are more concerned about solving problems than making others feel warm and fuzzy.

When you perceive rudeness, try to react calmly. If someone is really acting out, it is very likely that a senior person on the list or newsgroup or forum will call him or her on it. If that doesn't happen and you lose your temper, it is likely that the person you lose it at was behaving within the hacker community's norms and you will be considered at fault. This will hurt your chances of getting the information or help you want.

On the other hand, you will occasionally run across rudeness and posturing that is quite gratuitous. The flip-side of the above is that it is acceptable form to slam real offenders quite hard, dissecting their misbehaviour with a sharp verbal scalpel. Be very, very sure of your ground before you try this, however. The line between correcting an incivility and

starting a pointless flame war is thin enough that hackers themselves not infrequently blunder across it; if you are a newbie or an outsider, your chances of avoiding such a blunder are low. If you're after information rather than entertainment, it's better to keep your fingers off the keyboard than to risk this.

(Some people assert that many hackers have a mild form of autism or Asperger's Syndrome, and are actually missing some of the brain circuitry that lubricates `normal' human social interaction. This may or may not be true. If you are not a hacker yourself, it may help you cope if you think of us as brain-damaged. Go ahead. We won't care; we like being whatever it is we are, and generally have a healthy scepticism about clinical labels.)

In the next section, we'll talk about a different issue; the kind of `rudeness' you'll see when you misbehave.

On Not Reacting Like A Loser

Odds are, you'll screw up a few times, on hacker community forums -- in ways detailed in this article, or similar. And you'll be told exactly how you screwed up, possibly with colourful asides. In public.

When this happens, the worst thing you can do is whine about the experience, claim to have been verbally assaulted, demand apologies, scream, hold your breath, threaten lawsuits, complain to people's employers, leave the toilet seat up, etc. Instead, here's what you do:

Get over it. It's normal. In fact, it's healthy and appropriate.

Community standards do not maintain themselves: They're maintained by people actively applying them, visibly, in

public. Don't whine that all criticism should have been conveyed via private mail: That's not how it works. Nor is it useful to insist you've been personally insulted when someone comments that one of your claims was wrong, or that his views differ. Those are loser attitudes.

There have been hacker forums where, out of some misguided sense of hyper-courtesy, participants are banned from posting any fault-finding with another's posts, and told "Don't say anything if you're unwilling to help the user." The resulting departure of clueful participants to elsewhere causes them to descend into meaningless babble and become useless as technical forums.

Exaggeratedly "friendly" (in that fashion) or useful: Pick one.

Remember: When that hacker tells you that you've screwed up, and (no matter how gruffly) tells you not to do it again, he's acting out of concern for (1) you and (2) his community. It would be much easier for him to ignore you and filter you out of his life. If you can't manage to be grateful, at least have a little dignity, don't whine, and don't expect to be treated like a fragile doll just because you're a newcomer with a theatrically hypersensitive soul and delusions of entitlement.

Questions Not To Ask

Here are some classic stupid questions, and what hackers are thinking when they don't answer them.

Q: Where can I find program X?

A: The same place I'd find it, fool -- at the other end of a web search. Ghod, doesn't everybody know how to use Google yet?

Q: My {program, configuration, SQL statement} doesn't work

A: This is not a question, and I'm not interested in playing Twenty Questions to pry your actual question out of you — I have better things to do. On seeing something like this, my reaction is normally of one of the following:

- do you have anything else to add to that?
- oh, that's too bad, I hope you get it fixed.
- and this has exactly what to do with me?

Q: I'm having problems with my Windows machine. Can you help?

A: Yes. Throw out that Microsoft trash and install Linux.

Q: I'm having problems installing Linux or X. Can you help?

A: No. I'd need hands-on access to your machine to troubleshoot this. Go ask your local Linux user group for hands-on help. (You can find a list of user groups here: http://www.linux.org/groups/index.html.

Q: How can I crack root/steal channel-ops privileges/read someone's email?

A: You're a lowlife for wanting to do such things and a moron for asking a hacker to help you.

Good and Bad Questions

Finally, I'm going to illustrate how to ask questions in a smart way by example; pairs of questions about the same problem, one asked in a stupid way and one in a smart way.

Stupid: Where can I find out stuff about the Foonly Flurbamatic?

This question just begs for "STFW" as a reply.

Smart: I used Google to try to find "Foonly Flurbamatic 2600" on the Web, but I got no useful hits. Does anyone know where I can find programming information on this device?

This one has already SFTWed, and sounds like he might have a real problem.

Stupid: I can't get the code from project foo to compile. Why is it broken?

He assumes that somebody else screwed up. Arrogant of him.

Smart: The code from project foo doesn't compile under Nulix version 6.2. I've read the FAQ, but it doesn't have anything in it about Nulix-related problems. Here's a transcript of my compilation attempt; is it something I did?

He's specified the environment, he's read the FAQ, he's showing the error, and he's not assuming his problems are someone else's fault. This guy might be worth some attention.

Stupid: I'm having problems with my motherboard. Can anybody help?

J. Random Hacker's response to this is likely to be "Right. Do you need burping and diapering, too?" followed by a punch of the delete key.

Smart: I tried X, Y, and Z on the S2464 motherboard. When that didn't work, I tried A, B, and C. Note the curious symptom when I tried C. Obviously the florbish is grommicking, but the results aren't what one might expect. What are the usual causes of grommicking on MP motherboards? Anybody got ideas for more tests I can run to pin down the problem?

This person, on the other hand, seems worthy of an answer. He has exhibited problem-solving intelligence rather than waiting for an answer to drop from on high.

In the last question, notice the subtle but important difference between demanding "Give me an answer" and "Please help me figure out what additional diagnostics I can run to achieve enlightenment."

In fact, the form of that last question is closely based on a real incident that happened in August 2001 on the linux-kernel mailing list. I (Eric) was the one asking the question that time. I was seeing mysterious lockups on a Tyan S2464 motherboard. The list members supplied the critical information I needed to solve them.

By asking the question in the way I did, I gave people something to chew on; I made it easy and attractive for them to get involved. I demonstrated respect for my peers' ability and invited them to consult with me as a peer. I also demonstrated respect for the value of their time by telling them the blind alleys I had already run down.

Afterwards, when I thanked everyone and remarked how well the process had worked, an lkml member observed that he thought it had worked not because I'm a "name" on that list, but because I asked the question in the proper form.

We hackers are in some ways a very ruthless meritocracy; I'm certain he was right, and that if I had behaved like a sponge I would have been flamed or ignored no matter who I was. His suggestion that I write up the whole incident as an instruction to others led directly to the composition of this guide.

If You Can't Get an Answer

If you can't get an answer, please don't take it personally that we don't feel we can help you. Sometimes the members of the asked group may simply not know the answer. No response is not the same as being ignored, though admittedly it's hard to spot the difference from outside.

In general, simply re-posting your question is a bad idea. This will be seen as pointlessly annoying.

There are other sources of help you can go to, often sources better adapted to a novice's needs.

There are many online and local user groups who are enthusiasts about the software, even though they may never have written any software themselves. These groups often form so that people can help each other and help new users.

There are also plenty of commercial companies you can contract with for help, both large and small (Red Hat and LinuxCare are two of the best known; there are many others). Don't be dismayed at the idea of having to pay for a bit of help! After all, if your car engine blows a head gasket,

chances are, you will take it to a repair shop and pay to get it fixed. Even if the software didn't cost you anything, you can't expect that support will always come for free.

For popular software like Linux, there are at least 10000 users per developer. It's just not possible for one person to handle the support calls from over 10000 users. Remember that even if you have to pay for support, you are still paying much less than if you had to buy the software as well (and support for closed-source software is usually more expensive and less competent than support for open-source software).

SmoothWall FAQ and Troubleshooting Guide

Introduction

This document is intended to answer a series of the most commonly asked questions about SmoothWall and other related subjects. If you have any questions about SmoothWall, this document should be the first piece of documentation that you will need to refer to, as it contains the answers to the most commonly asked questions.

Please note that this is a “living” document, and as such, will be updated on a regular basis. If you do not find the answer to your question in the most recent version of this FAQ, and it has been asked a number of times previously, the chances are that it will end up in a later version of this document.

If you have not found an answer to your question here, in any of the other SmoothWall manuals (available at http://www.smoothwall.org) or in the admin web pages on-line help screens then you are welcome to ask the team. However please read the preface to this document thoroughly before proceeding and restrict your questioning to the SmoothWall mailing lists http://www.smoothwall.org and Internet Relay Chat (IRC) either available from the website or by pointing your IRC client to:

```
irc.smoothwall.org 6667
```

and join us on the #smoothwall channel. Please whatever you do, DO NOT e-mail the team directly, unless they specifically ask you to do so.

General Questions

SmoothWall

Q. What is SmoothWall?

A. SmoothWall is a specialised version of the popular Linux operating system that has been carefully designed, secured, and optimised in order to provide a network with all the functionality of a secure router and firewall at a fraction of the normal cost of doing so. Installing SmoothWall turns an everyday PC (typically an older system that has since become outdated and unable to cope with the demands of today's modern software) into a dedicated firewall that will protect a private network of computers from the dangers that are posed by connecting it to the Internet. SmoothWall not only protects your network from any unwanted attention from the Internet but also has the added advantage of rejuvenating and extending the useful life of an older PC system.

SmoothWall has been designed to be simple to install and operate, and yet remain secure and impenetrable. Installation is as simple as booting your PC with a SmoothWall CD, and configuring and maintaining the firewall as easy as pointing a web browser at the SmoothWall system.

Q. What version of Linux does SmoothWall use?

A. The Linux distribution that SmoothWall is based upon has its origins in the VA Linux customised Red Hat-

based Linux version. It has subsequently been optimised and heavily cut down in size so that all that remains is the bare minimum of core functionality that is required to provide a system that can operate securely as a network router and firewall. The version of the Linux kernel is 2.2.19.

Q. What are the main features of SmoothWall?

A. SmoothWall is a fully functional firewall that can be installed and run on any PC system from a 486 or upwards. It offers fault tolerance and the ability to audit and maintain the system from the convenience and ease of a web browser (such as Netscape or Internet Explorer) running on any client operating system. The easy to use administration system has been extensively tested with browsers running on Macintosh, Windows, and Linux platforms.

SmoothWall offers ease of use and much more - by providing a function known as NAT (Network Address Translation), it is possible to restrict access to the Internet to a single PC system and yet still enable all computers on the network to have full Internet capability. In addition, the inclusion of an internally-facing DHCP and DNS proxy server as part of the standard SmoothWall installation makes the configuration of the protected private network much simpler.

SmoothWall also has support for multiple network cards, Internet connectivity ranging from dial-up modem/ISDN through ADSL and leased line connections. It has a built in caching web proxy service, port forwarding capability, and an embedded

Java SSH secure shell to provide support for remote administration through a Java-enabled web browser.

Q. What kind of ISP connections does SmoothWall support?

A. SmoothWall offers protection for a network accessing the Internet using a wide variety of connection types, from dial-up modem access, through to broader band communications such as ISDN, ADSL, cable modems, and permanent leased line connectivity, and has been extensively tested for any weaknesses or compromises in security. No weakness have yet been found, proving that the "Smooth" in SmoothWall refers not only to the ease of installation and use, but also to the difficulty of finding any ”handholds” from which to gain unauthorised access to a network that is behind the protection of a SmoothWall firewall. Since the project’s inception SmoothWall has already proved very popular in a huge number of networks, ranging in size from small home networks to the realms of very large corporate networks.

Q. Where did the idea for SmoothWall come from?

A. SmoothWall was created from the need to service one specific requirement - that of protecting the way in which we work today - the computer network. Being able to provide a secure connection to the Internet was the key to this goal. Although the Linux kernel has enabled IP level security by means of the Ipchains functionality for some time, it can be hard, particularly for a non-technical or inexperienced user, to ensure that a system is properly secured against attack or any other

unauthorised use. Hence, one of the primary goals of the SmoothWall project was to provide a stable, simple to use, and yet totally secure, system that could protect a network of computers from attack from the Internet.

It quickly became apparent that the most suitable way of providing secure network connectivity was to create a cut down Linux distribution tailored specifically for servicing this need. Ideally this distribution would be able to make use of older and redundant hardware - in this case any PC with a 386 or greater chip, a network card, an IDE CD-ROM drive, a network card, and a small hard drive, capable of holding perhaps 60Mb or so of data. This specification covers a vast range of older hardware, but with systems at the lower end of this scale normally being too old to be able to boot from a CD there also had to be a method to enable installation of SmoothWall from a source other than CD.

It was surmised that if users could protect their networks by using a firewall such as SmoothWall by using easily available (and hence cheap) hardware they would do so. Considering the relative ease by which one can acquire an older PC system for very little outlay – for example a number of companies are often willing to donate (and write off) old and redundant hardware to their employees – this did not seem unreasonable to expect. This initial market research proved to be correct, and so development on SmoothWall began in earnest.

Initially SmoothWall offered support only for dial-up Internet connectivity, but it has since grown far beyond this and now offers secured Internet connectivity for a

range of broadband and permanent connections. For those users without a permanent connection, provision of secure dial on request connectivity has always been a key part of the project, with support for ISDN systems included as an early addition.

Q. How do I install SmoothWall?

A. Installation of SmoothWall is covered in detail in a separate document - the SmoothWall User Installation Guide. Please refer to this document for specific information regarding the installation of SmoothWall.

Q. How do I configure SmoothWall after installation?

A. Configuration of SmoothWall is covered in detail in a separate document – the SmoothWall Post-installation Configuration Guide. Please refer to this document for specific information regarding the configuration of SmoothWall.

Q. I don't understand some of the networking terms that the SmoothWall installation asks for – where can I find out more information?

A. The terminology of TCP/IP networking can be confusing to the newcomer, which is the reason why the SmoothWall team has provided a basic guide to TCP/IP networking to assist you in understanding more about the way SmoothWall works. The basics of TCP/IP networking, some more advanced networking concepts and a guide to basic network troubleshooting are covered in detail in a separate document – the SmoothWall Basic TCP/IP Networking Guide. Please

refer to this document for specific information regarding the network configuration of SmoothWall.

Q. Where can I find further information on SmoothWall?

A. The SmoothWall web site at http://www.smoothwall.org/ hosts a number of resources such as detailed guides that cover the installation and configuration of SmoothWall, this FAQ, and a basic guide to TCP/IP networking, which includes basic network troubleshooting. In addition, there are mailing lists and other

Q. What does SmoothWall cost?

A. Nothing - SmoothWall is freely available for use under the terms of the GNU Public Licence, a copy of which is can be found at *http://www.gnu.org/copyleft/gpl.html* or as part of the SmoothWall distribution. Though we ask you to make a charitable donation to The Dorothy Miles Cultural Centre which helps deaf and hearing people of all ages. More information is available at: http://www.smoothwall.org.

Q. Linux has a penguin. SUSE has a chameleon. Do you also have a mascot?

A. SmoothWall has a polar bear (aka "Smoothie") as its mascot. The more observant of you may have seen him poking his head up on the website, an in the title banner on the SmoothWall admin pages.

If you like you can download the buttons and banners from the SmoothWall website to use on your own website to link back to us, including Smoothie. You can get these down loads from:

http://www.smoothwall.org/gpl/about/evangelize.html

Firewalls

Q. What is a firewall?

A. A firewall is simply a system designed to prevent any unauthorised access to (or from) a private network of computer systems. This access control can be implemented by a hardware or software solution, or, as is often the case, a combination of both. Firewalls are frequently used to prevent access to a private network – such as, for example, your company Intranet - from unauthorised Internet users. All information (in the form of network traffic) entering or leaving such a private network passes first through the firewall, which examines the nature of this information, and depending on the rules that are part of the configuration of the firewall, allows this network traffic to either pass unimpeded, or to block it from going any further. As you might well expect, there are many different methods in which this overall goal can be achieved. SmoothWall has been designed as a packet level filter - that is, each and every packet of network traffic that passes through a SmoothWall firewall is inspected and is then either permitted to continue onwards, or is denied.

Q. Why do I need a firewall?

A. Well, you don't have to have one. You don't have to have a lock on your front door either. Firewalls offer a level of protection from other would be unauthorised

users of your network. There are a lot of people using the Internet these days, and some of them have no qualms about trying to get into your machine. If you don't want them there, you have two choices: a firewall or no connection to the Internet.

Q. How is SmoothWall different from other firewalls?

A. Some firewalls are software firewalls. They reside on the machine that is connected to the Internet, and act as a filter for information going in and out. The major drawback to a software firewall is that they have already connected to your box. It is like the difference between locking your front door, and locking your jewellery box. Both are meant to keep your jewels safe, but one is obviously more effective.

A hardware firewall (like a machine running SmoothWall) is between your network and the Internet. It forces anyone who wants to break in to have to go through an extra machine. The more work you make them do, the less likely they are going to want to spend the time on you. After all, the person down the street isn't protected at all. They are an easier target.

Support and communications

Q. I logged onto irc.smoothwall.org (port 6667) needing support and got my head bit off. How come?

A. This generally occurs when the question has been asked several thousand times before and the answer is available in the FAQ or Manual. Please check these first. The manual and FAQ are available at: http://www.smoothwall.org/

Q. I logged onto the #smoothwall irc channel, needing support, but the chat was about anything but SmoothWall. How come?

A. As well as people wanting support, #smoothwall is frequented by team members, who being dispersed around the globe use the channel to keep in touch, thus when not actively answering questions or supporting people, tend to chat amongst themselves. Though this may seem intimidating, if you had asked your question people would broken off to help.

Q. I was asked if I was @Home, which I confirmed, but then the support I was given didn't seem to make sense. Why?

A. You may well have confused the term '@Home' with 'at home'. @Home is an ISP who has some particular configuration issues. Hence once you mistakenly confirmed you were an @Home user, the support given was based on the known issues with @Home.

Q. I need support, but I don't have IRC, is there a mailing list?

A. Yes. The mailing lists can be found on the SmoothWall Website at: http://www.smoothwall.org/. Where you can also find a Java IRC client. Please keep the topic sensible - or you may find yourself being list banned.

Q. What's all that gobbledegook in the list at the bottom of a post?

A. That would be HTML from a user that hasn't worked out that lists are best posted to in plain text. Please only use plain text when posting to the lists.

Q. Can I post binaries to the list?

A. Please do not post binaries to the list. Upload it to somewhere and post a link!

Q. I have some thoughts on SmoothWall - are they welcomed?

A. Yes - certainly. If you are going to criticise, make sure it's constructive criticism. Don't just slate SmoothWall because a certain feature is missing or not to your liking! If you do have serious criticisms about please post them on the SmoothWall mailing lists or approach the members of the team directly, please do not use other public forums as this only causes hurt. Also don't get upset if your ideas are slated, the team have put many hours of their own time and expertise into developing SmoothWall, thus have a clear understanding of what it can, can't, should, shouldn't and will never do.

Q. My natural language isn't English. Do you have the manuals in my language?

A. Possibly. We rely on help from users to translate our software and manuals to foreign languages so the manual for your language may not be available. Perhaps you could help by translating or checking our documentation?

Q. What is everyone talking about when they are discussing "red", "green" and "orange"?

A. These are the types of networks that can exist off of a SmoothWall box. "Green" is the totally safe and trusted 'Private' network that you have your machine on.

"Red" is the Internet, with all the people who have no qualms about strolling through your personal files. "Orange" is an intermediary area. Partially safe, but outsiders can still access some of the services. If you are running an a public facing FTP server, Web server, it should be on the "orange" network (sometimes referred to as a DMZ or De-Militarised Zone).

Q. I really love SmoothWall, can I create a web site about it and how to install it and have copies of the files for download?

A. Firstly you have done the correct thing by asking this question, if you have already created and published a web site please take it down. Whilst we always welcome good publicity for SmoothWall, as with any product we like to have a degree control over what and how it is presented particularly such a rapidly evolving one. We therefore ask that you put a proposal forward to Richard Morrell and seek his permission before you proceed.

Functionality

Q. I don't understand exactly what Smoothwall is doing. I understand the principles of a firewall. What I don't understand is how I am able to return information back from, say, an external POP server. Why aren't the returning packets dropped? It bugs me when I don't understand how something works! Can anyone point me at a good explanation of SmoothWall / firewalls / networking?

A. In brief; the reason it works is due to the IP masquerade (sometimes known as NAT) employed.

This is a feature of the Linux Kernel (and other OSs) whereby the addresses of packets is re-written, such that a packet from a local network will appear to originate from the gateway. When the packet returns, the gateway then rewrites the packet again to give a local LAN address, and sends the packet back to the originating machine. It's much more complicated than this, though that's the basically how it works.

Masquerade is a method which allows you to prenst multiple machines as having a singular IP address. It is not without its limitations (E.G., as a local machine does not have a real IP address (as far as the internet is concerned) it cannot receive incoming connections directly). Of course, the added bonus is that since your desktop IP is effectively "masked", no-one on the outside can make connections to it. There are better answers and a description of Masqurading at:

http://www.linuxdoc.org/HOWTO/IP-Masquerade-HOWTO.html

Q. How do I use a SmoothWall box?

A. Smoothie sits between your private network and your internet connection. If you follow the installation and configuration instructions, your Smoothie will function automatically and you will not notice its existence. If you want to access your Smoothie's administration interface, just type the IP address or hostname that you gave the machine in the installation plus the redirection to either port 81 for http or port 445 for https in your browser window and this will remotely connect you to Smoothie. i.e.

http://198.162.1.1:81

or

```
https://smoothwall:445
```

Internet Service Providers

Telstra Bigpond Cable

Q. I subscribe to Telstra Bigpond Advance and I have to use special client to log into the service. Does SmoothWall support this?

A. No, not out of the box. However there is a mini-HowTo by Lucien Wells at http://www.users.bigpond.net.au/lwells/smoothwall/, which describes how to install the Linux BPALogin client onto a SmoothWall box. Please note that this is not a SmoothWall supported modification.

@Home

Q. I subscribe to @Home and I cannot get my Smoothie to get a DHCP lease from the @Home servers, why is this?

A. The @Home DHCP servers require that a client machine passes a specific hostname, before they issue a lease. To overcome this problem you should enter the hostname supplied by @Home in the DHCP hostname box on the Red NIC set-up page.

Homechoice

Q. I subscribe to homechoice in the UK which is fairly unique in the way it operates. It is an ADSL connection to a set-top box and then your PC connects to a serial port on the set-top box which is permanently connected to the internet and no dialing

or authentication is required as that is done by the set-top box already. Can SmoothWall be configured to handle this?

A. Yes, but the modification requires you to know how to operate the Linux editor VIM, information on which can be found on the internet at: http://www.vim.org.

1. Install using Green(+Orange)+Red(Modem or isdn)
2. Log onto your Smoothie as root either directly or from another machine on your internal network via ssh. Edit the dialer script which is located in /etc/ppp and comment out with a # the following lines:

```
ABORT '\\nBUSY\\r'                          \
ABORT '\\nNO ANSWER\\r'                     \
ABORT '\\nRINGING\\r\\n\\r\\nRINGING\\r' \
ABORT '\\nNO CARRIER\\r'                    \
```

Then comment out with a # all the lines from the next `if` command to just before:

```
$com =~ s/\n//g;
```

3. Open up the web admin pages go to the `dialup>ppp settings` page, set the idle timeout to 0, tick persistent connection, tick dod, dns on demand, authentication pap or chap and save your settings.
4. Reboot Smoothie and should see your modem connection light turn amber and stay on, fire up you favourite browser hopefully you're now connected.

NTL NTLworld Broadband, Telewest Blueyonder

Q. When I connect my Smoothie to my Cable Modem, I cannot connect to the internet, but I can if a connect my computer to my cable modem. What have I done wrong?

A. Essentially nothing, NTL have the cable modem configured to recognise only one client computer. Once the cable modem has learnt the MAC address of the first computer that talks to it, it will not respond to any other in any way. Thus if you swap one computer for another, the new computer will not work with the cable modem, because the new computer has a different MAC address to the old one. To reset the cable modem so that it will recognise the new computer, you must power it off and on again. Once the cable modem has rebooted and gone fully online again, reboot the newly connected computer so that it makes a DHCP request to the cable modem.

In some areas, it appears that resetting (or power cycling) the cable modem is not enough, and it is suggested that, before the first computer is disconnected from the cable modem, it should release its DHCP lease. In Windows 9x/ME, this is an option in the `winipcfg` command. In Windows 2000, type the command `ipconfig /release`. If even that is not enough, it might be necessary to wait for expiry of the current DHCP lease, and then reset the cable modem again, before it will recognise a new MAC address.

Q. When I connect my Smoothie to my Set Top Box, I cannot connect to the internet, but I can if a connect my computer to my Set Top. What have I done wrong?

A. The first time you connected your computer cable modem port of your Pace digital TV set top box, you will have launched a web browser and been redirected to a special registration site for customer MAC addresses.

Unfortunately the registration means that your set to box will only recognise the MAC address of your computer's NIC. There are two solutions to this problem either repeat the registration process, or swap the Red NIC in your Smoothie with the NIC in your computer.

This process of registering the client MAC address is quite separate from the registration of the cable modem HFC MAC address with the ISP.

Q. I have just upgraded to broadband, I have installed my Terajet 210 cable modem following the instructions supplied with it but my Smoothie does not configure its Red NIC using DHCP?

A. This is one of the few occasions where RTFM does not actually work, since Terayon have made an error in their manual. If you follow the instruction of powering up your Smoothie before the Cable Modem, Smoothie will be fully booted before the Cable Modem has finished testing and configuring itself, and thus Smoothie will have failed to pick up a DHCP lease from NTL. To overcome the problem power the modem first and allow it to test and configure itself

(this can take 5 minutes or more), and once the LEDs have stopped flashing power up your Smoothie.

Pre-installation

General

Q. What are the requirements for installing SmoothWall?

A. To successfully install and use a SmoothWall system to protect your network it is necessary to have a spare PC that can be made available for use as a dedicated SmoothWall system. Any data stored on the hard drive of this donor PC will be overwritten **without checking** as part of the installation, so it is imperative to back up any data that is considered to be valuable beforehand. This donor PC needs to be an Intel 486-compatible, Pentium, or higher (Pentium II, Pentium III), and it is recommended that a minimum of 16Mb of RAM is fitted for optimal performance.

Q. What checks can I do to make the SmoothWall install go smoothly?

A. Boot up and go into the BIOS, usually by pressing the *[DEL]* key. Disable all memory shadowing options. Disable Video BIOS and System BIOS cachable options. Disable Boot Sector virus checking (enable after the install). Make the bootup sequence A:,C; or CDROM,C: depending on the boot device. Check the first hard disk's parameters. If it shows more than 1024 cylinders, you may have a problem booting from LILO later. If you have ISA plug and play devices like some NICs, set up the device for no plug and play and choose non-conflicting IRQ, DMA and I/O ports. In the BIOS of a plug and play motherboard, reserve these

addresses in the legacy support section. If you have a caching IDE controller, like a Promise DC4030 VL, disable caching in that too. You can re-enable caching after the install. Check that the hard disk is the Master device on the Primary IDE channel. If you have an IDE CDROM, make this the Slave device on the Primary IDE channel. (Other combinations may work, but this is safest.)

Q. How do I install Smoothwall?

A. First, put the CDROM in your machine and try to boot from it. Some 486s won't let you do this. If yours does boot from the CDROM, SmoothWall will automatically start. Just follow the on screen directions. If it does not boot then you will need to make a boot floppy.

Note: SmoothWall will completely wipe the Hard Disk Drive of the installation machine.

Q. What if I don't have a CDROM on the machine I intend to use as the firewall. Can I still use Smoothwall?

A. You need to create both the boot and the drivers floppies. You can use these to install via FTP or HTTP. This is covered in the SmoothWall User Installation guide.

Hardware

Q. What sort of processor is required to run SmoothWall?

A. The size and speed of the processor that is required depends primarily on the amount of bandwidth that

will be protected by SmoothWall. For a modem or ISDN connection shared between a small number of computers a 486DX or low-end Pentium (P75 or P100) will be sufficient, but to process the network traffic generated by a larger number of users and to efficiently manage a larger bandwidth a faster processor will assist greatly.

The minimum specification that the SmoothWall system will theoretically operate on is a PC system with a 386 processor and 8Mb of RAM, but the lowest tested specification is that of a 486DX4 processor fitted with 8Mb RAM. Non-Intel (but Intel-compatible) processors have been found to work successfully, including processors from manufacturers such as Cyrix, AMD, and IBM.

Q. What size of hard disk is required to run SmoothWall?

A. The theoretical minimum is 60MB, however at this size you would not be able to successfully fun Smoothie's web proxy facility and would probably have problems with the disk being filled with logs and stalling your machine. 200MB is a comfortable size to use and on larger disk drives take the opportunity to use the space by increasing the size of the web proxy cache. You should however remember that old BIOs can have limitations as to the size of disk they can handle and that LILO (Smoothie's boot loader) is incompatible with disk managers such as On-Track. Thus on an older 486 machine you may well only be able to access the first 500MB of any hard disk drive you install.

Q. Do I need any other equipment?

A. You will need at least one network card (NIC), a keyboard (temporarily), a monitor (temporarily), a connection to the internet, a floppy drive and IDE CD-ROM. Once Smoothie is up and running, all regular maintenance can be performed remotely. Therefore the monitor is required for day-to-day operation. Once installation is complete the keyboard can also be dispensed with, however some BIOSes require a keyboard to boot properly. Most modern BIOSs allow you to disable the check for the keyboard on boot. However the keyboard can be useful for instigating a controlled shutdown should, for any reason, your Smoothie become inaccessible from the rest of your network, so you may choose to leave it connected. It is quicker to press *[Alt]+[Ctrl]+[Delete]* on Smootie's keyboard and wait for the beep, than it is to boot a client machine and shutdown using the Web Admin pages.

Q. Does SmoothWall supported SCSI devices?

A. No not in the current release. Whilst we recognise that there are SCSI users out there they are currently small in number and thus SCSI support is not high priority. Obtain a cheap IDE disk / CDROM.

Q. What NICs does SmoothWall support?

A. For a comprehensive list of the NICs and associated driver modules supported by SmoothWall please refer to the documents at http://www.linuxdocs.org, then review the /lib/modules/2.2.19/net directory on the SmoothWall CD-ROM or in the CD-ROM ISO file. In

this directory you will find all the NIC driver modules supplied with SmoothWall.

Q. What speed NICs should I put in my Smoothie 10Mb, 100Mb or 10/100Mb?

A. Since your internet connection is going to be much slower than 100Mb, typical DSL and broadband connections run at 512kb and 1Mbs, and the ethernet port on your Modem is typically a 10Mb connection, you only really need to purchase a 10Mb NIC for your Red NIC. Though if you money will extend that far you could purchase a 10/100Mb NIC. What you fit as your Green and Orange NICs, is as much dependant on configuration of each of these LANs as anything else, though if the Network on Green can run at 100Mb, there are advantages in using a card that can run at 100Mb if intend running Smoothie's web proxy.

Q. My network card doesn't have a driver for SmoothWall. How do I get it working?

A. Firstly check that the card isn't supported under a generic NE2000 driver. This should at least get you running. If you've still struck out, please post a message to the mailing list stating which manufacturer, model of card and FCC-ID (if present). Giving us as much information as possible will help us to help you. We can't promise an immediate fix - though it won't be forgotten.

Q. I have a Netgear FA311 NIC (the replacement for the FA310). Is this supported?

A. Not with the 2.2.x series of Linux kernels in this (0.9.9 or earlier) version of SmoothWall. It is supported by

the natsemi driver in the 2.4.x kernels. You can get a driver from the Netgear web site, or from Donald Becker's web site, but you will need to compile it etc.

Q. I'm thinking of installing SmoothWall on a laptop because of its small foot print and low power consumption. Do you support PCMCIA?

A. Whilst it has been recognised that a laptop Smoothie has many advantages the current version does not have PCMCIA support. However a little user effort it can be done. For further information look at:

http://libxg.free.fr/smoothwall/firewall.htm

Q. I've not got a QWERTY keyboard. Does SmoothWall support non QWERTY keyboards and how do I change my keyboard map?

A. During the installation process the Installation Manager will allow you to select the keyboard map which best suits your keyboard.

Q. Does SmoothWall work with v.92 modems?

A. Yes, there are users that have a Hayes Accura v.92 modems which work. Whether your ISP supports V.92 presently is a different matter.

Q. I'm using Token Ring for my network. Can I use SmoothWall?

A. No. Token Ring cards are not supported in this release.

Q. Does SmoothWall support winmodems as I have one on my motherboard?

A. No. SmoothWall only works with hardware modems.

Networking

Q. I have read all the manuals and all this IP, DHCP and Subnet business has me confused, I have none of this from my ISP. Could you just give me some numbers I can type into my SmoothWall box?

A. Given that you have never needed an IP address before your ISP is using some form of DHCP so you need to set your Red NIC (if you have one) to use DHCP. Set your Green NIC to use IP 192.168.1.1 the Netmask will be generated automatically.

Set up the DHCP on Green to use address range 192.168.1.100 to 192.168.1.200, and the primary DNS to be 198.162.1.1. You then need to set your client machines to use DHCP.

If you intend installing an Orange NIC, set this up with IP address 192.168.0.1. For machines on the Orange network you need to set them up with individual and unique IP addresses in the range 192.168.0.2 to 192.168.0.254.

Installation

Q. I've tried an installation and I need boot disks. How do I create them?

A. The generation of floppy disks is fully covered in the user installation guide.

Q. I'm doing an HTTP install. What do I type in for the URL?

A. Something like: 192.168.0.10/sw099/smoothwall.tgz You don't need the http:// at the beginning. The edit box has a limited line length, so don't put the smoothwall.tgz file too far down the directory hierarchy of your web server. You can usually set up a virtual directory like sw099 to point straight at the containing directory.

Q. What if I need more help at this stage?

A. There is an installation guide on the SmoothWall webs site at http://www.smoothwall.org and once you have read this manual, you can always ask questions on the e-list available at www.smoothwall.org . You can even obtain support via IRC at irc.smoothwall.org, channel #smoothwall, port 6667.

Q. I get error 0x10 when I try to install. What does this mean?

A. This is common problem and normally relates to bad install media, though occasionally it can be caused by old, faulty hardware such as your floppy drive or CD-

ROM. It is not untypical to go through 3 or 4 floppy disks before a good one can be found. If you have a linux machine use it to format and verify the floppy disk before you write the image. If you are using dos or windows, then carry out a full format before using `rawrite` or `rawwritewin`. Doing this will reduce the chances of having a bad floppy since the image writing routines do not verify what the write.

Q. The installation process hung without giving me any useful error messages. How can I find out what is going wrong?

A. During the installation process, *[ALT]+[F2]* shows the log of what's going on. Useful for diagnosing some kinds of error. *[ALT]+[F3]* puts you in a command prompt which is rarely useful. *[ALT]+[F1]* takes you back to the installation dialog.

Q. My display is only black and white and I cannot read many of the install dialogs. What can I do?

A. Temporarily put a colour display card in the PC for the install. You can put the black and white one back in once smoothwall is configured.

Q. Install cannot find my IDE CD-ROM. Where has it gone?

A. If you have an IDE CD-ROM, make this the Slave device on the Primary IDE channel.

Q. Install hangs when partitioning my hard disk. Why?

A. Your old IDE controller may not be supported by this Linux kernel, or you have Boot Sector Virus Checking enabled in the BIOS.

Q. I've downloaded the 0.9.9 ISO and tried installation on a 1GB harddisk. The install works fine up to the point where the system reboots. LILO starts and produces the message: LI 00 00 00 00 00..... which repeats forever.

A. Go into your BIOS and check your hard disk drive isn't set to LBA. If it is, set it to normal and then re-install. It could be that your hard disk drive contained another OS before. You could try the method below to 'clean' it.

Q. I've obtained a hard disk drive that I used in an SGI or Solaris box. I can't install SmoothWall on it. What should I do?

A. It may be that the hard disk drive needs properly cleaning so fdisk can utilise it. Try running:

```
dd if=/dev/zero of=/dev/hda count=1024
```

before fdisk to erase the partition table. You'll need a UNIX machine to do this. Of course it may just be that the hard disk drive is knackered.

Q. I think I need to enter some 'module parameters' how do I do this?

A. You'll probably have come across this with your network card. Example:

I have an NE2000 compatible card which needs parameters entering as not all it's settings could be auto detected. I know the IRQ and IO of the card (if you don't find these out first). I enter the parameters as:

```
ne io=0x320 irq=11
```

Q. I have two identical NICs in my SmoothWall. What module parameters would I enter for them?

A. Something similar to the following (assuming you has 2 off NE2000 NICs):

```
ne io=0x300,0x320 irq=10,11
```

The first card being io 0x300 irq 10. The second card being io 0x320, irq 11.

Q. How do I find out what my networks card's current setting are?

A. Most network cards come with a diagnostics program. Generally you require a bootable DOS disk. Boot from the DOS disk, change disks (if required) and run your diagnostics program. It should report (maybe in a sub menu) what the current settings are). If in doubt refer to your NIC's manual. If you have no manual or software, there a plenty of web sites which help in the identification of NICs and have links to other sites containing the necessary diagnostic and set-up programs.

Q. What information is sent back to smoothwall.org when I've installed? I feel like you've invaded my privacy - please justify this.

A. The InvBot reports back the following NON PERSONAL data and we don't actually have to tell anyone we're reporting back at all - but in the full edition the docs clearly state that;

"agreeing to use SmoothWall is a two way relationship. You as a company/individual will comply with the terms of the GNU General Public Licence and also respect the rights of all developers whose code lies within the project boundaries. SmoothWall requires you to automatically register the following information so that we can gauge our audience and also continue to develop SmoothWall for the future. SmoothWall will report back automatically the following data, no data concerning you or your civil liberties is infringed upon and no data collated will be used than for the purpose of continued development of this project."

Information collated:

- SmoothWall version installed
- CPU Vendor name
- CPU Model name
- CPU Megahertz
- RAM
- HDD size
- Connection type

- First two octets of IP address (to be reconciled against RIPE for geographical location information)."

This is ESSENTIAL information for us to gauge usage figures that are TOTALLY correct and also for us to see where we need to push effort. It also gives us control over where we need to put FTP servers, support effort and also where to push SmoothWall with user groups, local IT press and resources on the ground.

Registration is VOLUNTARY - Registration by definition gives us more information e.g totally in keeping with the guidelines laid down by the UK Data Protection act and all addenda to that act to date laid down by the UK Data Protection Registrar.

To see the VOLUNTARY registration form go to: http://www.smoothwall.org/

Post Installation Configuration

General

Q. I would like to install a {insert application} on my Smoothie, can you help me?

Ah, the grand daddy of all questions, quickly followed by and his brother:

Q. My Smoothie has a {insert suitably large number}GB hard disk drive and I would like to make use of it, can you help me?

A. Asking this question on the mailing list or IRC will inevitably result in the verbal equivalent of being hit round the head with a baseball bat. The answer is NO. SmoothWall turns a PC into a firewall device, and having installed the software you should no longer think of the box as being a PC. Similarly you should have thought about the size of the hard disk drive before you donated the PC. If you have plenty of spare disk capacity make it available to the web proxy server.

If you are still really desperate to install other packages on your Smoothie, well you can. It's up to you - SmoothWall is OpenSource and it's your box. However the SmoothWall Team will NOT support SmoothWalls which have been modified without official patches (we can't know what's on the system, how it was configured and what it's affecting).

Most Importantly: You may be opening yourself up to a security risk. After all you can change all the glass in windows in your house for paper sheets, but I doubt whether the police or your insurance company would look too kindly on it when your jewellery got stolen.

SmoothWall is a firewall – that is all it is designed to do. Not to serve news, Samba, NIS or anonymous FTP (or other such suggestions we've had).

For internal servers, put them on a machine on the GREEN network. For external servers, put them on a machine on the ORANGE DMZ network.

Q. I have installed SmoothWall on to my donor PC. However I can neither Admin Pages, nor ping my Smoothie. What is wrong?

A. There are a number of possibilities dependant on your installation.

1. Smoothie Red NIC connected to Client NIC:

   If you are using a direct connection rather than using a Hub you require a cross over cable rehter than a standard cable.

2. Smoothie with 2+ NICs:

   If your Smoothie has two or more NICs, it is quite possible that you have got the NIC mixed up when you plugged up the wiring. Simplychange the wiring and try again. If you are using a hub you can plug all the NICs the hub and try again. Once you have access remove the plug one at a time to ascertain which NIC is which.

Software

Q. I have installed SmoothWall and now I can't use {insert name of software product} across the internet what have I done wrong?

A. Basically you have done nothing wrong, it is what has not been done that is the problem. In a situation like this your first port of call should be the software manuals and, or the web site associated with the product.

There are three basic reasons why a product will not function through a firewall:

1. The software will not function in association with SmoothWall.
2. The software needs to be reconfigured to run with a firewall.
3. Various ports need to be enabled and forwarded to allow the software to converse through the firewall.

This leads neatly on to the next question:

Q. I have searched for SmoothWall on the {insert name of software package} web site, can recover zero entries, what can I do now?

A. Just as we do not mention every software product that you could conceivably use in conjunction with your Smoothie, it is highly unlikely that other companies will list SmoothWall as a product on their website, far more likely that they will have general information about firewalls and ports, so search again using terms like 'firewall', 'ethernet' and 'port'

Q. I can't DCC through SmoothWall using my mIRC client. Why?

A. We've found this occurs quite a bit. It's not a SmoothWall related issue - more a protocol / differing clients issue. Help is at hand in the FAQ at the mIRC web site.

Port forwarding and External Access

Q. I want to allow external users access to port {n} on a machine on the Green network of my SmoothWall box how do I do this?

A. This is best explained with a worked example:

We wish to forward external users to our web server, which has IP 192.168.1.200 on our Green Network. Our web server runs on port 8080. Our SmoothWall Green IP is 192.168.1.1

On the *services>port forwarding* web admin page enter the following rule which will set up the forwarding instruction:

Protocol	Source Port	Destination IP	Destination Port
TCP	80	192.168.1.200	8080

On the *services>external service access* web admin page enter the following rule which will open up port 80.

Protocol	Source	Destination Port
TCP	ALL	80

Q. I want to allow external users access to port {n} on a machine on the Orange network (DMZ) of my SmoothWall box how do I do this?

A. This is best explained with a worked example:

We wish to forward external users to our web server, which has IP 192.168.0.200 and is on our Orange Network. Our web server runs on port 80. Our SmoothWall Green IP is 192.168.1.1 and the Orange IP is 192.168.0.1.

On the *services>port forwarding* web admin page enter the following rule which will set up the forwarding instruction:

Protocol	Source Port	Destination IP	Destination Port
TCP	80	192.168.0.200	80

On the *services>external service access* web admin page enter the following rule which will open up port 80.

Protocol	Source	Destination Port
TCP	ALL	80

Q. I want to allow a machine on the Orange network (DMZ) access to port {n} on a machine on the Green network of my SmoothWall box. How do I do this?

A. This is best explained with a worked example:

We wish to allow our webmail server which has IP 192.168.0.200 and is on our Orange network access to port 110 on our mail server which has IP 192.168.1.200 and is on our Green Network. Our SmoothWall Green IP is 192.168.1.1 and the Orange IP is 192.168.0.1.

On the *services>dmz pinholes* web admin page enter the following rule which will set up the pinhole:

Protocol	Source IP	Destination IP	Destination Port
TCP	192.168.0.200	192.168.1.200	110

Q I have set up a SmoothWall box at a remote office, and I wish to be able to manage from my office. I don't want to make the Web Admin generally accessible but I have a dynamic IP. How do I do this?

A. This is best explained with a worked example:

We wish to access SmoothWall admin externally but wish to restrict access to the IP network range used by our ISP for their DHCP servers which is 123.145.789.0

On the *services>external service access* web admin page enter the following rule which will open up port 445.

Protocol	Source	Destination Port
TCP	123.456.789.0/255.255.255.0	445

Q I have set up a server behind my SmoothWall, and configured the necessary port forwarding. However I only want to make the server assessable to a single fixed IP. How do I do this?

A. This is best explained with a worked example:

We wish to access a pop server located behind our SmoothWall, the necessary port forward has been configured, but we only which to make the server accessible to our remote office whose IP is 123.145.789.9

On the *services>external service access* web admin page enter the following rule which will open up port 110.

Protocol	Source	Destination Port
TCP	123.456.789.9	110

Internet Access

Q. I am using trying to use Dial on Demand, but my modem never seems to hang-up. What have I done wrong?

A. First of all check to make all the applications on your Green Network machines are correctly configured, particularly for thing like web proxy servers. Secondly if you are Microsoft Windows on these machines, they may well be producing WINS DNS calls on a 10 minute cycle, which Smoothie's DNS is trying to resolve. Since the default idle time is 15 minutes, and the WINS DNS calls occur on a 10 minute cycle the modem will never drop. To stop these DNS calls from causing external traffic insert add entry into your Smoothie's `/etc/hosts` file, consisting of a spoof IP address and the hostname:

```
<workgroup>.<domain name>
```

Where <workgroup> is the workgroup name set-up on your clients and <domain name> is the domain name you entered on your Smoothie.

Q. My modem or external ISDN TA responds to AT commands and needs an INIT string. Where do I put it?

A. The modem INIT can be entered in the appropriate box on the *dialup>modem* web admin pages.

Q. My ISP disconnects me after a certain period of time. How to I make SmoothWall reconnect me automatically?

A. Click the 'Persistent Connection' checkbox which is found with your ISP account settings on the *dialup>ppp* web admin page.

Q. Why doesn't SmoothWall connect to my ISP successfully?

A. It's probable that your ISP has a 'non-standard' setup. Post your problem to the mailing list and we'll determine the problem. Alternatively, if you know your ISP works fine in your other Linux boxes, look at the logs and compare them with a successful dial from your other Linux machine. If your still stuck, send us both the logs.

Q. When I try to dial via SmoothWall it gets as far as "dialling" and then all that happens then is the screen flashes. In the logs the following error message appears several times, what causes this?

```
18:11:47 kernel isdn: HiSax,ch0 cause: E001B
```

A. From the ISDN4Linux FAQ:
http://www.isdn4linux.de/faq/

7.7 trouble_e001b: I get an error message with "cause: E001B"?

This is a very popular error and means (see man isdn_cause): euro ISDN (E), location user (00), and out of order (1b). Taken together means that the driver either can't get a layer 1 connect (cable problem, hardware error, hidden hardware conflict - see section hardware), or it can't get a layer 2 connect (wrong configuration: no Euro ISDN, no

automatic TEI supported, point-to-point BRI instead of multi-device - see section config).

Whilst ISDN cause codes are cryptic and not interpreted in the logs, the answer is close to hand. "man isdn_cause" gives you all the reasons.

Security

General

Q. Help, I have just downloaded and run Leaktest from grc.com and my Smoothie has failed.

A. Calm down, think logically and look at what Leaktest does. Leaktest is a classic FUD spreader, first of all read what the Leaktest web page actually says.

> *'LeakTest pretends to be an FTP client application which attempts to connect to port 21 (FTP) of one of our servers within the grc.com domain.'*
>
> ©2001 by Gibson Research Corporation

Well knock me down with a feather, SmoothWall actually allowed a computer on the Green network running an FTP client to connect to an FTP server on the Internet. If it had not, you would probably be reading this document to find out why you could not connect to FTP servers through Smoothwall.

If you are really worried about Viruses, Worms, Trojans etc. Then you should do the following:

1. Invest in a decent Anti-virus software package and keep it up to date.
2. Monitor your application suppliers for security bulletins and install patches and fixes as soon as they are released.

3. Take and retain regular backups of critical applications and data that are stored on your machines.
4. Have a strict policy about opening e-mails with attachments, and information on portable media from any source.

You should be doing all the above anyway. If you are still paranoid, then the simplest answer is not to your private network to the outside world or to accept any software unless guaranteed virus free by the manufacturer. Failing that get rid of all your computers and go back to pen and ink.

Q. Is SmoothWall 100% watertight? Is it true it's un-hackable?

A. We try to make SmoothWall as watertight as possible. You should never assume that ANY firewall is 100% hack proof. To date we don't believe that SmoothWall has been hacked.

Q. I have a security worry - where can I go for help?

A. Please send an e-mail stating your concern to:

security@smoothwall.org

We'll try and get back to you ASAP, and we would prefer you do this rather placing your concerns in the public arena before we have been able to assess whether your concern is a real security risk and before we have managed to produce an update.

Q. I'm interested in network / computer security. Is there any useful sites or information out there?

A. Sure - as long as you use the information to protect, not gain access to unauthorised systems you should check out:

- http://www.insecure.org
- http://www.securityfocus.com
- http://www.hackers.com
- http://www.cerias.purdue.edu/coast/hotlist/index.html

Q. I used one of those internet firewall testing sites. It said that my ICMP port was open. Is this a problem?

A. While some people would like to close that port as well, ICMP (Ping) was consciously left open to allow you to run diagnostics on your firewall. All a hacker can get from a ping is that your machine exists and is alive. Having this port open is not a security hole.

Q. Is it safe to allow external automated sites to scan my network / firewall?

A. No it isnt. This is the easiest way for an attacker to harvest IP addresses with the owner's consent. Once they have the IP they will often send back bogus reports and have a nice database of insecure boxes to play with There are many tools available that will allow you to test your own set-up.

Q. I did a nmap port scan of my SmoothWall and found that 1025 is open. Help?

A. Port 1025 on Smoothie is dnrd, the dns proxy / cache. This port is needed to receive DNS info from external DNS servers. You cannot block this without killing DNS proxy functionality. dnrd runs as non root and is chroot in an empty directory.

Q. I'm worrying about how SSH is configured in Smoothie by default : - which algorithm is used for encryption ?

A. It depends on the client - it can vary from 3DES to Blowfish to CAST.

Q. Is the whole session encrypted or just the authentication?

A. The whole session is encrypted - keys are traded before you are challenged for your password. SSH is very well done and reasonably secure - more than enough for the purposes of remotely connecting to your Smoothie and doing some remote admin.

Q. Why is Smoothie showing my ports are open? For example, a remote UDP scan from http://scan.sygatetech.com showed that I have ports 137 (NetBIOS-NS), 138 (NetBIOS-DGM), and 139 (NetBIOS) open. Are the scans from this site accurate? How do I turn off these ports?

A. Some users of cable modems may find that they have those netbois ports "open". They appear almost as if the cable company / manufacturer has set up a "honey pot" on those ports from the outside. This may vary with different manufacturers or suppliers.

VPN

Q. Can you direct me to some documentation about how to setup VPN functionality with Smoothwall 0.9.9?

A. There isn't a HOWTO on VPN specifically for Smoothie, but the Freeswan site: http://www.freeswan.org has a reasonably easy to follow guide for setting up a simple VPN. Static IP to Static IP, with a simple shared secret is "easy" to implement.

Logs

Q. I use NTL / Virgin as my ISP and I'm getting some repetitive logs similar to that below. What/why is this?

TIME	CHAIN	IFACE	PROTO	SOURCE	PORT	DEST	PORT
15:23:11	Input	eth1	UDP	62.253.65.217	65535	244.0.0.1	65535

A. This is multicast traffic originating from somewhere in your ISP.

Client Configuration

General

The final stage in getting your network protected is to configure your desktop client systems to use the SmoothWall system as their gateway to the Internet.

The simplest method is to use SmoothWall's built-in DHCP server. Once you have configured the DHCP server settings on the SmoothWall system you must enable DHCP support in the network configuration of the operating system of your desktop clients.

When the desktop systems are next powered up the SmoothWall DHCP server will assign each system an IP address, provide details of DNS servers, and will set the default gateway to be that of the SmoothWall system.

If you choose not to use the SmoothWall DHCP server and instead intend to use static IP addresses for your client systems, you must change the network settings of these systems to use the SmoothWall system's IP address for the DNS server and default gateway addresses.

Microsoft Windows 9X

In order to change your Windows network settings, first double click on the *Network* icon in the Windows Control Panel (you can also right-click on the Network Neighbourhood icon on the desktop and select the *Properties* menu item).



Figure 1 - Selecting the Network icon in Control Panel.

This will bring up the Network properties of the computer. Please note that the examples shown below are of a basic configuration that can use TCP/IP over the LAN. Established networks may have other network protocols such as NetBEUI or IPX/SPX already in situ. If the TCP/IP protocol is not is not already installed you will have to install it and bind it to the network card. Do this by selecting the *[Add]* button, followed by *Protocol* and select the *[OK]* button.

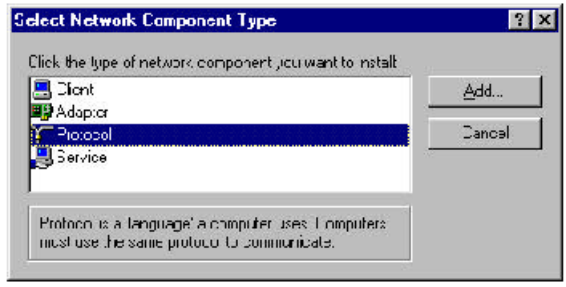


Figure 2 – Selecting the Protocol to add.

At this point select *Microsoft* from the list of manufacturers and *TCP/IP* from the list of available protocols.

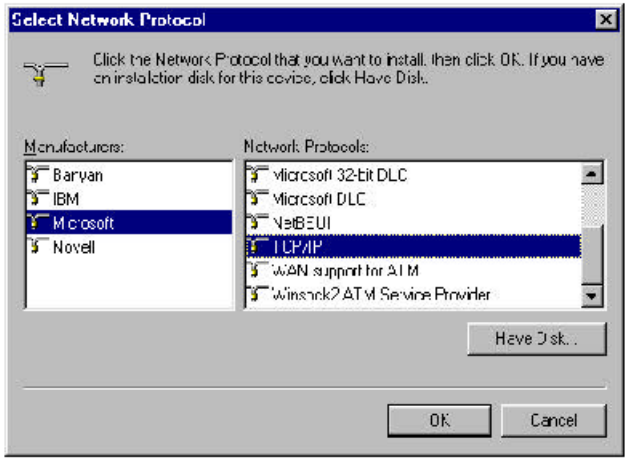


Figure 3 – Adding the Microsoft TCP/IP protocol implementation.

It is likely at this point that you will be prompted for your Windows CD, and then the system will wish to be rebooted. When you have successfully installed the TCP/IP protocol and it is bound to your network card you can configure the TCP/IP network properties. Highlight *TCP/IP* from the network configuration window and select the *[Properties]* button.

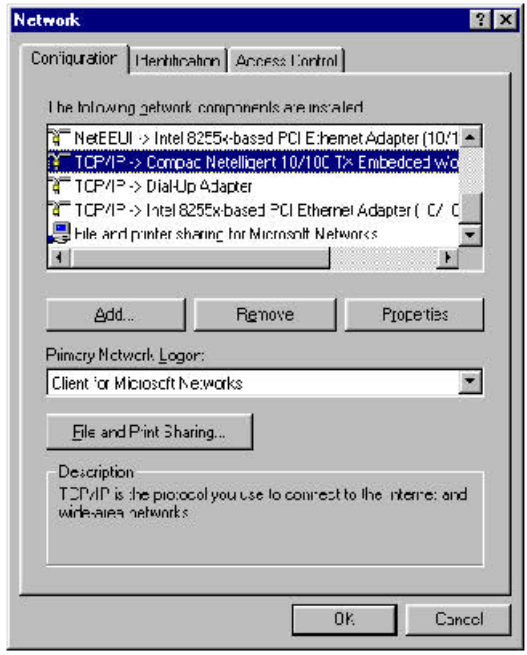


Figure 4 – Select the TCP/IP properties from the network configuration screen.

Your TCP/IP properties will look something like this. If you wish to use the SmoothWall DHCP server select the *Obtain an IP address automatically* option, and if you wish to use static IP addresses fill in the *IP Address* and *Subnet Mask* with the settings required for your network.

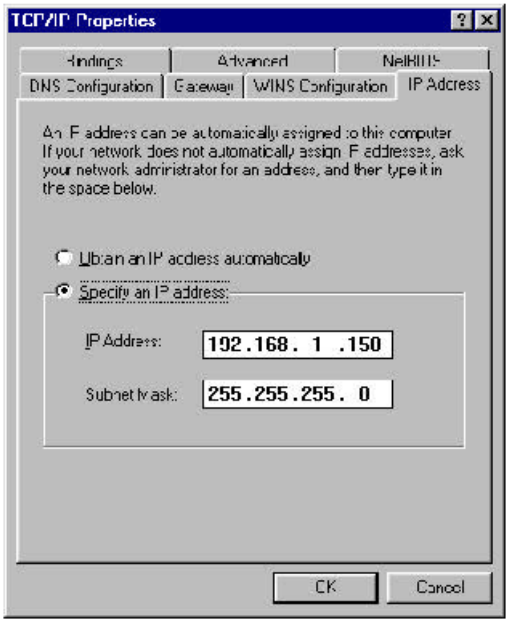


Figure 5 – TCP/IP properties.

If you are using static addresses and not the SmoothWall DHCP server to supply further information about your network to each desktop client you will have to add the necessary details yourself. Select the *Gateway* tab and enter the IP address of your SmoothWall system. Also select the *DNS Configuration* tab, enter the IP address of your SmoothWall system and select *Add*.

All other details of network configuration should be left at their default values.

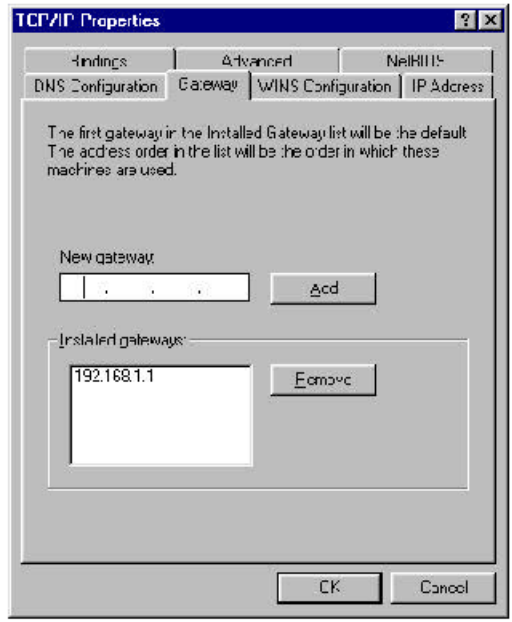


Figure 6 – Setting the gateway address.

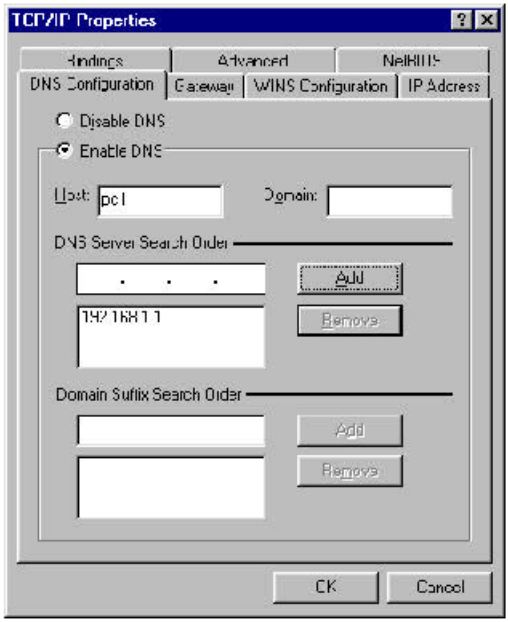


Figure 7 – Setting the DNS properties.

When the network configuration is completed click the *OK* button in the Network Properties window. You could be asked for the Windows CD, and then after loading anything necessary, the system will wish to reboot. When the PC comes back up again you should be able to communicate over the network using TCP/IP.

In order to test this you can use the *ping* command as detailed in the SmoothWall Basic TCP/IP Networking Guide – instructions for doing so are given in the section on network troubleshooting.

If this is successful you can now connect to the Internet from this PC over the LAN and be completely secure in doing so.

Glossary

DHCP

Dynamic Host Configuration Protocol - a protocol for assigning dynamic IP addresses to devices on a network. With dynamic addressing, a device can have a different IP address every time it connects to the network. In some systems, the device's IP address can even change while it is still connected. DHCP also supports a mix of static and dynamic IP addresses.

Dynamic addressing simplifies network administration as the software keeps track of IP addresses rather than requiring an administrator to manage the task. This means that a new computer can be added to a network without the hassle of manually assigning it a unique IP address.

FUD

Fear Uncertainty and Doubt – a classic salesman’s method of scaring in no buying into something, by using misleading information.